

Dimension of the Four Subspaces

Some Fast Check:

- ① Column space $C(A)$ and Row space $C(A^T)$ both have dimension $\boxed{r}$ (The rank of A)
- ② For a r rank matrix with n columns, the number of free variables would be $\boxed{n-r}$.
 $\hookrightarrow$ Also the number of special solution.
- ii) If a matrix has no dependent column, then there exists no free variables either.
 Thus the null space of that matrix is zero vector only.
- iii) If a matrix has $n-r$ dependent columns, then the number of basis vectors of the null space are $\boxed{n-r}$.
 the dimension of the null space is $\boxed{n-r}$

Fundamental Theorem of Linear Algebra

Part 1:

✱ The column space and row space both have dimension r .

✱ The nullspaces have dimensions $(n-r)$ and $(m-r)$

Before we dive into the class notes, let us go through the dimension of the four subspaces:-

Fundamental Space	Dimension	Basis Vector
Column Space of $A \Rightarrow C(A)$	r	Pivot Columns
Null Space of $A \Rightarrow N(A)$	$n-r$	Special Solution
Row space of $A \Rightarrow C(A^T)$	r	Pivot Rows of $A / R(A)$
Left Nullspace of $A \Rightarrow N(A^T)$	$m-r$	Free Rows of E

Four Subspaces

(A is $m \times n$)

Column Space $C(A) \Rightarrow$ in $\mathbb{R}^n$

Null Space $N(A) \Rightarrow$ in $\mathbb{R}^n$

Row Space $C(A^T) \Rightarrow$ All combination of the rows of A in $\mathbb{R}^m$

Row space $C(A^T)$

$\Rightarrow$ All combination of the rows of A in $\mathbb{R}^n$
 $\Rightarrow$ All combination of the columns of A^T

Left null space $N(A^T) \Rightarrow$ in $\mathbb{R}^m$

A Visual Representation of the spaces

Row Space

$$\dim C(A) = \text{rank } A$$

basis = pivot rows of $A/R(A)$

Column Space

$$\dim C(A) = \text{rank } A$$

basis = pivot columns

Null Space

$$\dim N(A) = n - r$$

basis = Special Solution

Left Null Space

$$\dim N(A^T) = m - r$$

basis = Free rows of E

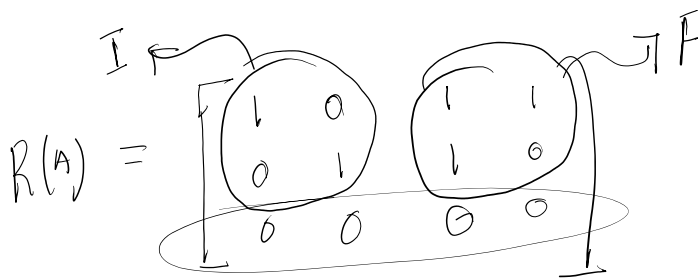
Example:

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$$

$$\hookrightarrow = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad [\text{Row Reduction}]$$

$$= \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$I \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} F$$



Here, $C(R) \neq C(A)$

⊛ Before we dive in to the next chapter let's try to interpret, why row space is orthogonal to null space.

$Ax = 0$ $\rightarrow$ Set of vectors that get mapped to zero

$\leadsto$ Applying A to x results in the zero vector

$\leadsto$ Span of the rows of matrix A

if A has m rows then $A = \begin{bmatrix} \pi_1 \\ \pi_2 \\ \vdots \\ \pi_m \end{bmatrix}$

$$\text{So } Ax = \begin{bmatrix} \pi_1 \\ \pi_2 \\ \vdots \\ \pi_m \end{bmatrix} x = 0 \rightarrow \left. \begin{array}{l} \pi_1 x = 0 \\ \pi_2 x = 0 \\ \pi_3 x = 0 \\ \vdots \\ \pi_m x = 0 \end{array} \right\}$$

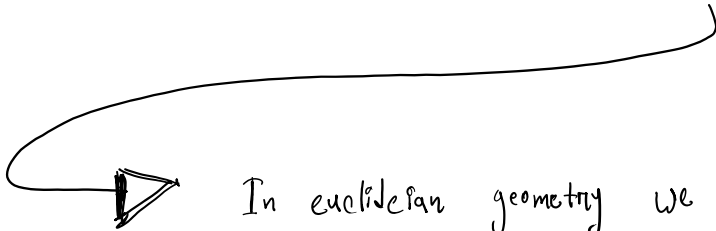
This tells us that for every vector π_i of A the dot product with any null space vector x is zero. This means that x does not contribute to the row space at all.

$\rightarrow$ Since x does not contribute to the row space in any way, it must be completely independent of it

Thus the null space must be entirely separate from the row space in terms of vector contributions

⊗ $\vec{v} \cdot \vec{x} = 0 \rightarrow$ this is basically the definition of perpendicularity in vector spaces

$\hookrightarrow$ How so?



In euclidean geometry we are used to thinking of perpendicularity as a 90° angle. But in linear algebra, perpendicularity is defined by the dot product being zero.

This will be explained again later on